Online Resource 1: Proposed Empirical Designs

Supplement to *Grounding Without Corrective Control: Truth-Tracking Profiles for Large Language Models*

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11th August 2026

Journal: *Synthese*

1 Purpose

This resource specifies three component tests proposed in the accompanying article: a crossed route-by-task design, a comparison that varies checking-route independence, and a temporal test of live answerability. The designs don't jointly distinguish the decomposition from an unrestricted effective-coverage description. Their purpose is to test whether a decomposition fixed before the outcomes are observed supports selective transfer, diagnosis, and intervention. No results or outcome magnitudes are reported.

2 A crossed route-by-task design

The worked example uses clinical questions. One item set asks for recommendations recoverable from guidelines or institutional records. A paired set uses the same disease area and answer format and requires a patient-specific lab value, observed sign, dose calculation, or intervention-sensitive causal judgement. The prediction is a retrieval gain on the record-recoverable set and little transfer to the patient-specific set unless the needed measurement or tool route is supplied. Retrieval producing the same gain on both sets would count against the proposed distinction between routes. The design parallels validity argumentation in measurement: inferences from performance are licensed relative to a purpose and tested by varying method while holding the target construct fixed (Kane, 2013; Messick, 1989).

Matching the sets on difficulty cues isn't sufficient. An intervention that supplies a missing lab value also makes the item easier, so a difficulty explanation and a route explanation both predict a gain. Equating measured baseline accuracy weakens the simplest difficulty account without eliminating differences in discrimination, ceiling structure, response variance, latent subskills, or treatment-effect heterogeneity. Matching on noisy estimates also invites regression to the mean. Use baseline performance as a covariate and design diagnostic, not as evidence that difficulty has been removed.

The primary design is crossed rather than merely matched. Build item families in which the relevant route can be supplied or withheld within closely related items, then cross route availability against task requirement. Estimate the interaction in a hierarchical model with varying effects for items, domains, prompt templates, and systems. Declare in advance the smallest interaction that would matter, report its interval, and run reasonable alternatives for item matching, scoring, and exclusion rather than relying on one sample.

In the single-pair implementation, the comparison uses two named chat arrangements that differ only in whether retrieval over a fixed corpus is available, with the prompt template held constant. The estimate is conditional on that pair; variation across model families and deployment setups lies outside the estimand and belongs to a multi-system extension. The population is a clinical item set split into record-recoverable and patient-specific halves. Raters classify the items from their text without seeing system output, and the classification is frozen before the first run.

The primary outcome is substantive correctness of the recommendation, scored blind, which applies to both halves of the item set. Whether verbatim anchors resolve is secondary and applies only within the record-dependent half. Making anchor resolution primary would manufacture the interaction because patient-specific items have no corpus anchors to resolve.

The worked prediction is that the retrieval gain on record-recoverable items exceeds the gain on patient-specific items by at least ten percentage points. Ten is illustrative, not derived. A defensible threshold needs a decision basis: the cost of a wrong recommendation, the cost of retrieval latency, and what a deployment would do differently at nine points rather than eleven. Fix the value before outcomes are observed so that the prediction can fail.

The interval is interpreted against that threshold rather than zero. Concentration above ten points supports the worked claim; concentration near zero or on the wrong side counts against it. An interval extending from below zero to well above ten describes an uninformative experiment, not evidence either way. The threshold isn't revised after the effect becomes visible.

A stronger implementation crosses both interventions with both task types. Retrieval should selectively help record-dependent items; a measurement tool should selectively help items requiring measurement. This double dissociation tests the proposed route carving, though a relevance-sensitive information account can predict the same qualitative interaction.

3 Checking-route independence

Compare two checkers: one from the generator's lineage and the other trained and calibrated on separate data and feedback. Give them the same candidate output and task evidence supplied from outside the checkers. Match standalone accuracy, confidence, latency, response format, and revision policy as closely as possible. Measure detection and correction conditional on generator error, especially under perturbations expected to produce common-mode failure.

For this comparison, input coverage is measured before either checker produces a task-specific signal. Checker outputs are downstream consequences of the architecture, not part of baseline input coverage. If coverage is measured after a checker responds, the predicted advantage is already a conditional-coverage effect.

The decomposition predicts an advantage for the checker with lower conditional error overlap. Calibration history matters only insofar as it predicts that overlap; if admissible perturbations produce no difference, the decomposition has no reason to privilege one history. An input-coverage account predicts no advantage once externally supplied task information and standalone accuracy are matched. A conditional-coverage account can predict one by representing the joint error structure, but that prediction incorporates the relation that checking-route independence names.

4 A temporal test of live answerability

Give two arrangements the same target information at the outset, one holding a frozen pasted record and the other holding a live route to its source. In the update arm, correct the source record and ask a follow-up whose answer turns on the correction. In the control arm, leave the record unchanged. The live route should track the update in the first arm and match the frozen copy in the second. Figure S1 summarizes the design.

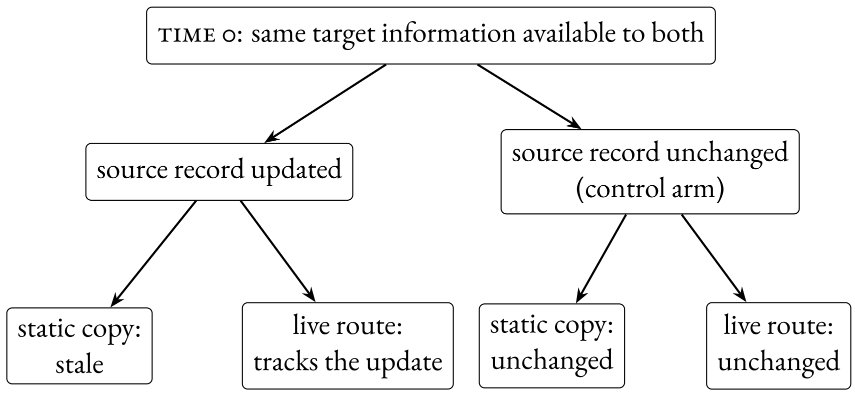

Figure S1: A temporal test of live answerability. At time 0, both arrangements have the same target information. In the update arm, the source record is corrected: the static copy stays stale while the live route transmits the correction. In the control arm, neither should change. No outcome magnitudes are implied.

A dynamic coverage account predicts the same pattern because the arrangements receive different information at follow-up. The manipulation operationalizes live answerability and isolates updateability from initial content equivalence; it doesn't adjudicate between the decomposition and dynamic coverage.

5 Manipulation checks and failure conditions

Failure of a declared route-specific interaction or update effect counts against the decomposition only after confirming that the focal route was invoked and that its output reached production in time to affect the answer. A stale index, failed tool call, or ignored checker signal shows that the intended manipulation wasn't instantiated. Once those checks pass, failure of the declared contrast is evidence that the proposed route carving or corrective feature has been misdescribed.

References

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